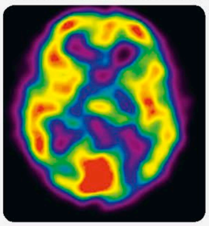
HOW CAN WE STUDY THE EFFECTS OF SPIRITUALITY?

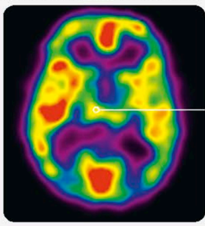
Neuroscientists are now studying the brain during spiritual states, with fascinating findings. American neuroscientist Dr. Andrew Newberg from the Marcus Institute of Integrative Health, for example, uses neuroimaging to understand higher spiritual states, including spiritually based meditative states such as Samadhi, deep prayer practices, and some drug-induced spiritual experiences (see below).

Four common brain patterns in spiritual experiences

Dr. Newberg has compared the brain at rest and during transcendent spiritual experiences, including Samadhi, to identify specific brain activity patterns associated with spirituality.

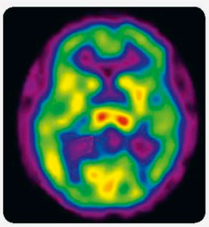
AT REST | **DURING SPIRITUAL EXPERIENCE** | **INTERPRETATION**

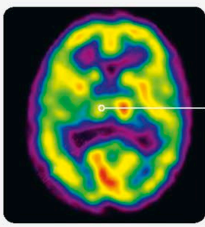




Increased activity in the limbic system

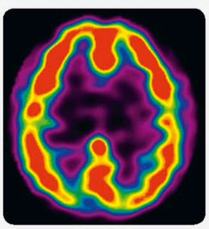
INTENSITY
Increased activity in the limbic system may account for the intense emotional states that people often feel during spiritual experiences. This increase is also likely to make such experiences memorable and life-changing.

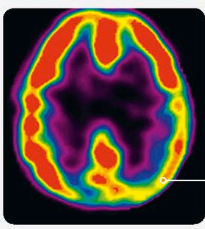




Decreased activity in the left thalamus

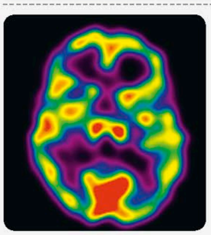
CLARITY
The thalamus is a relay center that helps us integrate sensory information to construct our sense of reality. Decreased activity here may result in a sense of increased clarity.

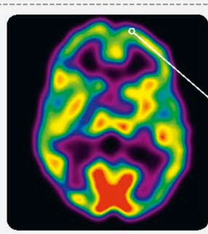




Decreased activity in the posterior parietal lobe

UNITY
The posterior parietal lobe is in charge of spatial orientation. A decrease in activity here may reduce the feeling of being physically separate from what is around us, creating a sense of unity and a lack of boundaries.





Decreased activity in the frontal cortex

SURRENDERING OF SELF
Though many meditation practices increase activity in the frontal cortex due to the increase in concentration and regulation, spiritual states such as Samadhi may turn off the frontal cortex, the seat of the will, leading to a sense of surrendering to what is.